
Compositional Concept Synthesis via CLIP-Guided Primitive Stitching

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Abstract

In this paper we explore the task of creating a composite image given a natural language prompt from a set of given primitives, such as combining “person” and “umbrella” to form “a person under an umbrella.” Each primitive is represented as a visual element, and a composition function arranges them spatially to form a complete scene. To guide this composition, we use CLIP to evaluate how well the rasterized image of the composed scene aligns with a natural language description of the target concept. The optimization objective is to maximize the cosine similarity between the image embedding and the text embedding produced by CLIP.

1. Introduction

Our work focuses on the synthesis of complex composite image by stitching together a set of predefined primitives. Each target concept is associated with a collection of k primitives, which are arranged to maximize the cosine similarity between the generated image embedding and the corresponding text embedding produced by CLIP. Initially, we implemented this task using the `pgpelib` library, where each concept was directly mapped to its corresponding primitives.

Early experiments revealed that our initial optimization methods were insufficient, leading us to explore alternative strategies. We discovered that the Covariance Matrix Adaptation (CMA) algorithm performed slightly better.

Despite the improvements provided by CMA, the performance of our method was hindered by a challenging loss function landscape. The loss, defined as the negative cosine similarity, exhibited many local maxima, which impeded the optimization process from converging to the global optimum. To address this, we modified the loss function by

incorporating an additional component designed to smooth the landscape. This adjustment aimed to mitigate the effect of local maxima, thereby facilitating more robust convergence.

While this enhanced approach yielded to better results, it also introduced certain limitations that will be discussed in detail in the following sections.

2. Related Work

We based our work on the paper *Modern Evolution Strategies for Creativity: Fitting Concrete Images and Abstract Concepts* (Tian & Ha, 2022) (ES-CLIP).

ES-CLIP presents a method that leverages evolutionary strategies to optimize image generation based on CLIP’s joint embedding space. An element in the *population* is composed of a set of overlapping triangles eachone with its own vertex position and color. These features are optimized to maximize the cosine similarity between the CLIP embedding of the triangles rendered on the canvas and the CLIP embedding of a target prompt.

Inspired by this work, we want to do something similar by composing visual scenes from a set of semantic primitives, instead of triangles.

3. Method

In order to stitch together k primitives to form a scene, we need to define a few components: the parameter to be optimized and the optimization strategy.

3.1. Parameters & Pipeline

We represent the parameters as a matrix of shape $(k, 5)$ and each entry is bounded to be in the $[0, 1]$ range. Each row corresponds to one primitive and the five parameters describe its spatial attributes.

- **x, y**: the position of the top-right corner of the primitive with respect to the canvas size;
- **scale**: the relative size of the primitive (i.e. a percentage of the full primitive size);

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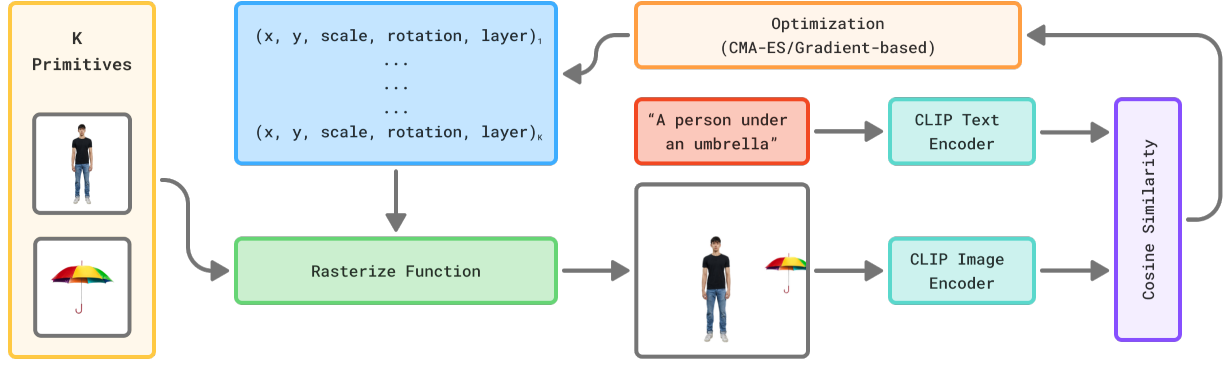


Figure 1. The whole pipeline of our method. We want the k (in this case 2) primitives shown in the yellow box, to be rasterized to have a similar CLIP embedding as the prompt, shown in the red box.

- **rotation**: the angle of rotation $r \in [0, 365]$ degrees normalized;
- **layer**: the rendering order (i.e., which primitives appear in front). We order the rasterization process, such that primitives with smaller layer parameter will be rasterized first, hence be in the back.

We created a `rasterize` function, which uses these parameters and the primitives to compose the scene and rasterizes it into an image. This image is then fed into CLIP’s image encoder to compute its embedding. The cosine similarity between this image embedding and the text embedding of the target prompt serves as the objective function. An overview of this pipeline is shown in Figure 1.

3.2. PGPE Optimization

The first algorithm we tried is PGPE (Probabilistic Gradient Population Evolution) (Toklu et al., 2020), the one used in the ES-CLIP paper. This algorithm is a gradient-free optimization method that uses a population of candidate solutions to explore the search space. This method did not yield good results as in the implementation of the ES-CLIP paper. We believe that one of the problems here are the **sparse rewards**, we will discuss this further in the next section.

3.3. CMA-ES Optimization

To overcome the limitations of PGPE, we decided to use a more advanced Evolution Strategy implementation, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Hansen & Ostermeier, 1996). Compared to PGPE, CMA-ES uses a more sophisticated sampling algorithm, which prioritize directions where improvement has been observed updating a multivariate Gaussian distribution. Unfortunately even if it produces slightly better results compared to PGPE, this method does not bring a significant improvement.

3.4. Gradient-Based Approach

In parallel, we also experimented with a gradient-based optimization method. We created a new rasterization function which is differentiable and has some limitations: rotating the primitives was pretty hard, so we decided to implement it only if a preliminary test would have been promising. This function allowed us to compute gradients of the cosine similarity loss with respect to the primitive parameters and apply gradient descent to improve the layout. Unfortunately it did not yield good results as well.

4. Problems

In this section we will discuss the problems we encountered during the optimization process and why the ES methods didn’t work as in the original paper.

4.1. The Sparse Rewards Problem

We hypothesized that one of the reasons our approaches didn’t work is the fact that in our problem the solutions have very **sparse rewards**. It means that sampled points in the surroundings of a certain point in the solution space yields inconsistent rewards values and not a clear gradient direction. If the best solution is far from the sampled points, the algorithm will not be able to reach it, since intermediate solutions do not provide an incremental improvement in the reward. Therefore the loss landscape is very noisy, often inconsistent, has many local minimum and it has a lot of “flat regions” (i.e., regions where the loss does not change much). It means it is difficult for the algorithm to find a path to reach the optimal solution without running into a local minimum.

4.2. The CLIP problem

We decided to investigate the sparse rewards problem further and we wondered why our loss landscape was so noisy.

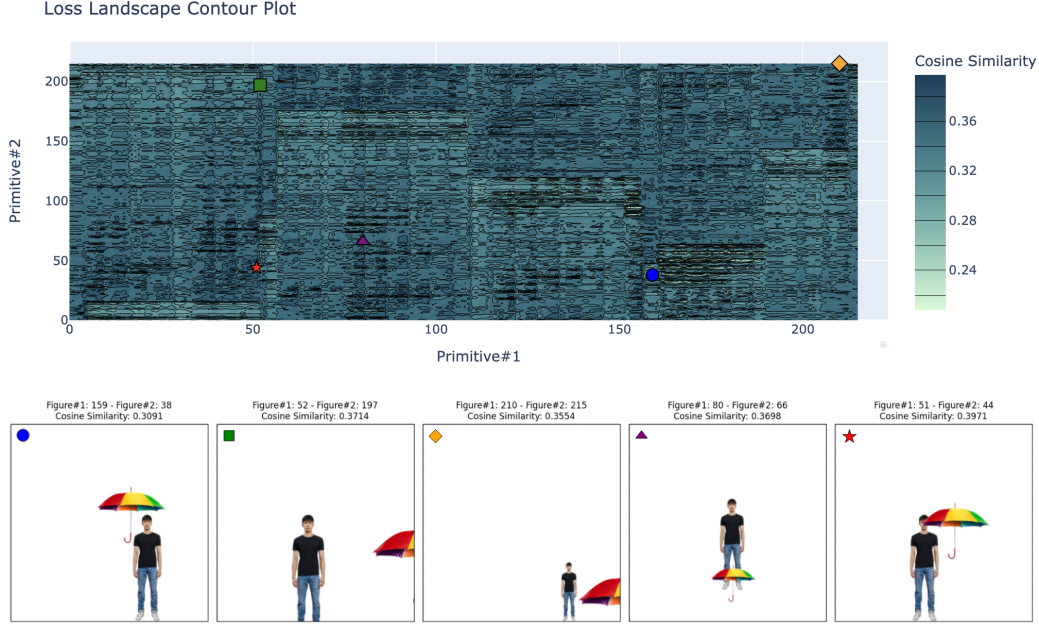


Figure 2. Loss landscape of the function for the prompt "A person under an umbrella".

We found out that the CLIP model is not very good at distinguishing between similar images. Moreover in some cases the CLIP model is not able to determine which image is better than the other, even if one is the manually created one and the other is a meaningless composition of primitives as we can see an example in Figure 3, the prompt used for the similarity computation is "A car on the road". The ground truth image (a) has a CLIP similarity score of 0.2675 while the generated one (b) has a score of 0.2755

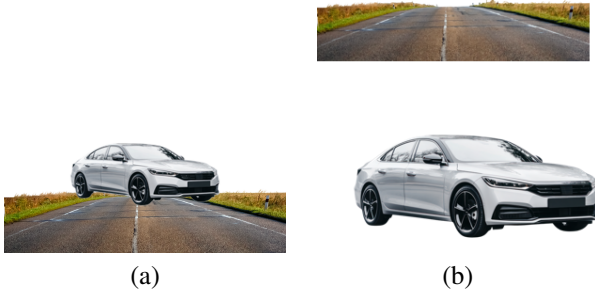


Figure 3. Examples demonstrating CLIP's inconsistency.

- (a) The ground truth image
- (b) The generated one

4.3. More on the loss landscape

After the observation we did before we decided to explore the loss function's landscape. In Figure 2 plotted the contour plot of the loss landscape. We choose a simple example "A person under an umbrella", that has

two primitives: *person* and *umbrella*. We embedded every combination of $(x, y, scale)$ parameters with values 0.3, 0.38, 0.46, 0.54, 0.62, 0.7 for both primitives, while keeping the rotation and layer fixed. We confirmed empirically that the loss function is very noisy, has many local optima and flat regions. To give a further proof of the loss flatness zones, we computed the *Coefficient of variation* of the values we computed in Figure 2. The coefficient of variation is a measure of the relative variability of a distribution, a low value indicates that the values are close to the mean. The Coefficient of variation is 0.0583, it means that the values are very close to the mean and that the loss function is very flat.

5. Results

As we can see in Figure 4, the image with the red background is the best result we obtained meanwhile the image with the red caption in the one with the highest similarity. The image with the highest fitness is not the best one confirming our hypothesis.

6. Conclusions

Despite implementing several optimization strategies, including PGPE, CMA-ES, and gradient-based methods, our results fell short of expectations. Our investigation revealed significant challenges that impeded the optimization process:

- sparse rewards in the solution space making conver-

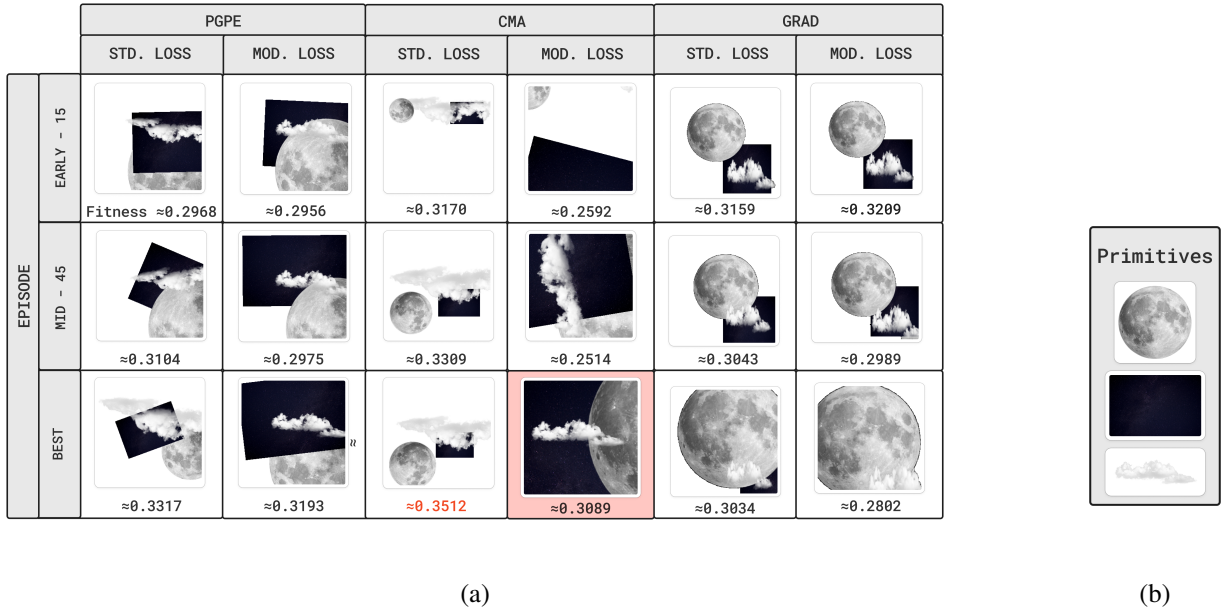


Figure 4. Results for each method with both the original and the modified loss function.

gence difficult

- inconsistencies in CLIP’s ability to discriminate between semantically meaningful and random arrangements
- a highly non-convex loss landscape with numerous local optima and flat regions.

These findings suggest that future work should focus on developing more robust optimization algorithms specifically designed to navigate sparse reward landscapes. Additionally, incorporating auxiliary reward signals or hierarchical optimization approaches might help guide the search process more effectively. Pre-training on simpler compositions before attempting more complex scenes could also provide a more informed initialization for the optimization process.

References

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